

#klimatapotrebuje

Klima Kompas Platform

Komunálne Voľby 2026

Capabilities Document — Version 5.0 (LOCKED)

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Confidential — Internal Use Only

CAPABILITIES INDEX — V5

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CAP-06	Human Review & Approval Dashboard MVP P0
CAP-07	Scoring Engine & Badge Assignment MVP P0
CAP-08	Manual Evidence & Override System MVP P0

Data Flow	Description
CAP-08 → CAP-07	Manual evidence entry triggers score recalculation (covers all pillars — SLOVÁ and SKUTKY — wherever AI agents are unavailable or fail)
CAP-02 → CAP-07	AI program/questionnaire scores trigger recalculation
CAP-05-LITE → CAP-02	Questionnaire submission triggers AI analysis
CAP-07 → CAP-06	Computed scores placed in review queue
CAP-06 → CAP-01	Approved scores published to public site
CAP-03 → CAP-07	Automated voting scores (Phase 2)
CAP-04 → CAP-07	Social scores (Phase 2)

CAP-01**Public Candidate Scorecard Website**Phase
MVP | P0**GENERAL SECTION****GS: Objective**

Website-first. Build the display layer before the data pipeline. A working URL with fake candidate cards is the NGO stakeholder pitch. Then fill it with real data.

GS: MVP Feature Scope

- Homepage: interactive SVG map of Slovakia's 8 krajov
- Region view: 2 candidate rows (Predseda kraju + Primátor); badge + name + party
- Candidate detail: overall %, badge, SLOVÁ bar (40%), SKUTKY bar (60%), source citations list
- Grey badge: single visual; sub-type text in detail view only
- Methodology page: formula, thresholds, Climate Relevance Framework, normalisation worked example
- Legal disclaimer on every card: "Toto hodnotenie nie je odporúčaním na hlasovanie."
- Social media note on every card: "Hodnotenie sociálnych sietí bude doplnené v ďalšej fáze."

GS: Post-MVP (Phase 2)

- Candidate comparison (side-by-side)
- Evidence density indicator ("Skóre na základe X zdrojov")
- Score version history / appeals display
- OG image generation (dynamic social share cards)

GS: Assumptions

- Scorecards published only after NGO human approval.
- All 16 Phase 1 positions always visible, even if Grey.
- Read-only for public — no user accounts, no PII collected.
- Appeals handled manually outside the system in MVP.
- OG images produced by NGO manually in MVP.
- Hosting: Lovable's hosting through development; production hosting NGO decision.

USE CASES**UC: Voter Views Scorecard**

1. Voter opens homepage — SVG map of Slovakia with 8 krajov.
2. Taps kraj — 2 candidate rows with badge + name + party/independent.
3. Taps candidate — detail: %, badge, SLOVÁ bar, SKUTKY bar, citations.
4. Taps any citation — source URL opens in new tab.
5. Taps "Metodológia" — methodology page.

UC: Grey Badge

- List view: grey badge + "Nedostatok dát" — identical visual for all sub-types.

- Detail view: sub-type explanation text (NO_DATA / REFUSED / NEW_CANDIDATE / LOW_CONFIDENCE).

NON-FUNCTIONAL REQUIREMENTS

NFR: Performance

Low priority. No high traffic expected. Pre-computed scores with basic caching is sufficient. No performance hard requirements in MVP.

NFR: Accessibility

Nice-to-have. Implement accessibility improvements incidentally during build (e.g. aria-labels when writing components). No dedicated audit in MVP.

NFR: Security

- Public routes: read-only, no auth, no PII
- Admin routes: Supabase Auth with MFA
- Analytics: aggregate page views only — Plausible or Fathom preferred

SC: Technical Notes

- Slovakia SVG map: embedded SVG — no external mapping API dependency
- URL pattern: /kandidat/[region]/[position]/[slug]
- Candidate state machine drives all conditional rendering

CAP-02

AI Program & Questionnaire Analysis Agent

Phase
MVP | P1

GENERAL SECTION

GS: Objective

Automate SLOVÁ pillar scoring for programs (Carter method) and questionnaire responses (NRSR +1/–1 rubric). Fastest AI win — cleanest inputs.

GS: Two Scoring Methods

Method A — Programs: Carter et al.

Classify every program sentence as pro-climate, anti-climate, or neutral. Compute: `pro_sentence_pct - anti_sentence_pct` = raw score. Normalise using Carter global caps: min -35.30, max +17.31. Formula: $\min(100, \max(0, (\text{raw} + 35.30) / 52.61 \times 100))$. Specificity sub-score applied within pro-climate sentences: 0=vague, 1=specific, 2=specific+timeframe. Adjusts effective `pro_sentence_pct` upward for high-specificity programs. Local relevance filter: named local infrastructure = full weight; national-level response to local question = 0.5 weight. AI does initial pass. NGO reviewers verify minimum 20% sample. If agreement < 80%, full manual review triggered.

Method B — Questionnaires: NRSR +1/–1 Rubric

+1 per specific measure (defined solution + purpose OR timeframe). -1 per anti-climate commitment. Local relevance filter applied. Normalise: $\min(100, \max(0, \text{raw} / 54 \times 100))$. 54 = NRSR cap (PS 2023).

GS: Assumptions

- PDF text extraction first; Claude Vision OCR fallback for image-only PDFs.
- Questionnaire non-response: `questionnaire_score` = 0. Not null. Does not alone trigger Grey.
- Prompt injection guard: system prompt instructs model to ignore instructions inside analysed documents.
- If agent fails: reviewer enters scores via CAP-08. No blocking dependency.
- All evidence items include `climate_relevance_tier`. For programs: Tier 1 if the sentence itself uses environmental language (emisie, klíma, životné prostredie, measurable env. target); Tier 2 if the environmental link is real but the program does not state it — AI must flag these and include a `reviewer_note` explaining the link; Tier 3 if no credible environmental connection.

USE CASES

UC: Analyse Program

6. Trigger: `analyzeProgram(candidateId)`
7. Fetch document; text extraction; Claude Vision OCR fallback if needed
8. Carter prompt → per-sentence classifications with tier tags and specificity scores
9. Compute raw score → normalise → store to programs table with full evidence schema
10. Insert into `review_queue`; candidate state → ANALYZED

UC: Analyse Questionnaire Response

11. Trigger: questionnaire response webhook from CAP-05
12. NRSR rubric prompt → specific measures with tier tags + local relevance scores
13. Compute raw total → normalise → store evidence items
14. Re-trigger CAP-07 recalculation

Alternative Flows

- Fetch fails: retry ×2; mark fetch_failed; alert reviewer; manual entry via CAP-08.
- Malformed Claude output: retry temperature=0; if still fails, mark parse_error; reviewer enters manually.

Mandatory evidence schema — every score point, every pillar, every source

- source_type: council_vote | resolution | initiative | program | questionnaire | social_post | manual_entry
- url: required; internal storage URL if physical document scanned; N/A only with admin approval + second reviewer
- date_accessed: ISO timestamp
- citation_text: verbatim excerpt ≤280 chars
- confidence: float 0.0–1.0
- climate_relevance_tier: 1 (explicit — source states environmental connection) | 2 (implicit — reviewer documents environmental connection) | 3 (excluded — no credible environmental connection; not scored; requires secondary reviewer to reclassify)
- reviewer_note: required if tier=2 (must document environmental connection), if tier=3, if score adjusted, or if no-URL exception

NON-FUNCTIONAL REQUIREMENTS

NFR: Performance

- < 120 seconds per candidate end-to-end
- 16 candidates in < 1 hour in single batch
- Exponential backoff; max 5 concurrent Claude API requests

NFR: Security

- Claude API key in Supabase Vault — never in source code or client
- Prompt injection guard in system prompt

CAP-03

Voting Record Parser Agent

Phase
MVP pilot |
Phase 2 full |
P2

GENERAL SECTION

GS: Objective

Automate extraction of climate-relevant votes from zastupiteľstvo minutes. Phase 2 full rollout. CAP-08 manual entry is the MVP primary path.

GS: Climate Relevance Tier Classification for Votes

Tier is determined by how the candidate or council framed the vote — not by the topic of the vote. The same transport vote can be Tier 1, Tier 2, or Tier 3.

Tier	Classification Rule Agent Instruction
Tier 1 — Explicit	The vote proposal, council resolution, or accompanying document contains explicit environmental language: emisie, klíma, životné prostredie, obnoviteľné zdroje, CO2, or a measurable environmental target. Agent instruction: scan the vote proposal document (not just the agenda item title) for environmental keywords. If found, classify Tier 1. Cite the exact passage.
Tier 2 — Implicit	The vote has a real and demonstrable environmental effect, but the source document does not frame it in environmental terms. The agent must generate a reviewer_note explaining the environmental connection. Agent instruction: classify Tier 2 when a vote involves transport modal shift, urban green space, waste reduction, air quality improvement, or energy efficiency — but the proposal lacks explicit environmental language. Write reviewer_note: "Environmental connection: [specific reason]." Flag for human confirmation.
Tier 3 — Excluded	No credible environmental connection in the proposal or any related document. General infrastructure, economic, or administrative votes. Agent instruction: classify Tier 3 when no environmental keyword is present AND no obvious environmental effect exists. Store item but exclude from scoring. Flag for secondary reviewer if borderline.

Keyword pre-filter (efficiency pass before LLM): scan for any of these before sending to Claude API — emisie, CO2, klíma, obnoviteľné, solárne, SEAP, elektromobil, cyklodoprava, MHD, zeleň, park, biodiverzita, ovzdušie, odpad, energia. If none found: classify Tier 3 directly without LLM call (cost saving). If any found: send to Claude API for full classification.

GS: Assumptions

- Scope: 2022–2026 electoral term plenary minutes only.
- Source URLs seeded manually by NGO admin per jurisdiction.
- Fuzzy name matching: Jaro-Winkler ≥ 0.90.
- Tier 3 items stored but not scored; flagged for secondary reviewer.
- All items enter review queue regardless of confidence.

USE CASES

UC: Phase 2 Automated Pipeline

15. Admin seeds base URL per jurisdiction.
16. Agent crawls, downloads all PDFs in scope.
17. Text extraction; Claude Vision OCR fallback.
18. Keyword pre-filter → tier pre-classification.
19. Claude API confirms tier and vote direction (pro/anti).
20. Fuzzy name matching links vote to candidate.
21. Evidence items created with full schema including tier.
22. CAP-07 recalculates.

NFR: Fallback

- If CAP-03 unavailable: CAP-08 manual entry covers voting records. No blocking dependency.

CAP-04

Social Media Analysis Agent

Phase
Phase 2 | P3

GENERAL SECTION

GS: Objective

Automate SLOVÁ social sub-score (10% of total). Phase 2 only. Social = 0 and reweighted in MVP.

GS: Scoring — 3-Axis Classification (NOT Carter)

- Axis 1: Environmental relevance (0=none, 1=tangential, 2=direct)
- Axis 2: Local specificity (0=national/abstract, 1=regional, 2=specific named local issue)
- Axis 3: Concreteness (0=sentiment, 1=describes problem, 2=specific action/commitment)

Post score = average of 3 axes. Recency weight: posts in final 30 days count 0.5× to reduce last-minute greenwashing. Normalisation caps: TBD before Phase 2 implementation based on pilot data.

GS: Scope

- Public accounts only: Facebook (primary), Instagram, X/Twitter. 12-month window.
- Engagement metrics explicitly excluded. Content only.
- Misinformation flagged for human review — never auto-confirmed.

CAP-05

Questionnaire — LITE (MVP P1) + FULL (Phase 2)

Phase
MVP P1 lite |
Phase 2 full

GENERAL SECTION

GS: MVP Objective

Accept and store questionnaire responses for AI analysis. Dispatch and tracking managed manually by NGO in MVP.

GS: MVP Scope

- Generate unique UUID submission link per candidate.
- Serve submission form at that link.
- Store submitted response in questionnaire_responses table.
- Trigger CAP-02 analysis on submission.
- Show in CAP-06: submitted / not submitted (binary).

GS: MVP Out-of-Scope → Phase 2

- Automated reminder emails (Day 7, Day 14) — NGO sends manually.
- Full state machine (SENT → REMINDER_1 → REMINDER_2 → NO_RESPONSE).
- Empty Chair public display logic.
- RESPONDED_INCOMPLETE vs RESPONDED_COMPLETE distinction.

GS: Questionnaire Structure (both phases)

8 questions targeting local competencies: transport, energy, green space, air quality, waste, water, climate strategy, EU funding. 6 open-ended, 2 binary (Áno/Nie/Neviem + optional justification). Target response time: 45 minutes.

USE CASES

UC: MVP — Accept Submission

23. NGO admin generates unique link per candidate via CAP-06.
24. NGO sends link manually by email.
25. Candidate submits form; response stored; webhook triggers CAP-02.
26. CAP-06 shows: "Response received" for this candidate.

NON-FUNCTIONAL REQUIREMENTS

NFR: Audit Trail

- Server-side submission timestamp.
- Response SHA-256 hash stored at submission.
- Dispatch tracking: NGO manages in spreadsheet in MVP; imported to system in Phase 2.

CAP-06

Human Review & Approval Dashboard

Phase
MVP | P0

GENERAL SECTION

GS: Objective

NGO operational control centre: review evidence, approve scorecards, generate questionnaire links, manage candidate pipeline. Nothing goes public without passing through here.

GS: MVP Feature Scope

- Review queue: candidates sorted by state
- Candidate review panel: evidence items with Accept / Adjust (note required) / Flag actions
- Approve for publication: transitions state to PUBLISHED
- Generate questionnaire submission link per candidate
- Candidate management: add/edit name, photo, party, region, position
- Current computed score shown per pillar for review context

GS: MVP Out-of-Scope → Phase 2

- Bulk CSV/JSON import
- Appeals SLA tracking and in-system resolution
- Score version history UI

USE CASES

UC: Review and Approve

27. Reviewer logs in; sees review queue.
28. Opens candidate — split panel: evidence list (left) + source viewer (right).
29. Per evidence item: Accept (A) / Adjust (E — note required) / Flag (F — admin escalation).
30. All items actioned → "Approve for Publication."
31. CAP-07 publishes to public site.

UC: Generate Questionnaire Link

32. Admin opens candidate record.
33. Clicks "Generate questionnaire link."
34. UUID link displayed for admin to copy and email manually.
35. Admin marks "Link sent" via manual checkbox in CAP-06.

NON-FUNCTIONAL REQUIREMENTS

NFR: Usability

- Split-panel layout: evidence list + source viewer
- Keyboard shortcuts: Accept (A), Adjust (E), Flag (F), Next (N)

- Progress indicator: X/16 candidates approved
- Desktop-first

NFR: Audit (MVP — minimal)

- Every approval and adjustment logged: reviewer_id + timestamp
- Full immutable audit trail and export: Phase 2

NFR: Security

- Supabase Auth with MFA
- Roles: researcher (review + approve), admin (override + publish + candidate management)

CAP-07

Scoring Engine & Badge Assignment

Phase
MVP | P0

GENERAL SECTION

GS: Objective

Backbone: compute KLIMA_SCORE, assign badge, version-control results. Pure arithmetic — no LLM at score time. Runs in < 1 second.

BUSINESS REQUIREMENTS

BR: Formula — Version 1.0 (LOCKED)

KLIMA_SCORE = (SLOVÁ × 0.40) + (SKUTKY × 0.60)

SLOVÁ (40% of total):

- MVP: SLOVÁ = (program_norm × 0.50) + (questionnaire_norm × 0.50)
- Phase 2: SLOVÁ = (program_norm × 0.375) + (questionnaire_norm × 0.375) + (social_norm × 0.25)

SKUTKY (60% of total):

- SKUTKY = (votes_norm × 0.417) + (actions_norm × 0.583)
- votes_norm: min(100, max(0, (raw_votes + n×2) / (n×4) × 100)) where n = climate votes in term
- actions_norm: min(100, max(0, (raw_actions + 10) / 25 × 100))

Normalisation — all components:

- program_norm: min(100, max(0, (raw + 35.30) / 52.61 × 100)) [Carter caps]
- questionnaire_norm: min(100, max(0, raw / 54 × 100)) [NRSR cap]
- actions_norm: min(100, max(0, (raw + 10) / 25 × 100)) [designed cap]
- votes_norm: min(100, max(0, (raw + n×2) / (n×4) × 100)) [jurisdictional]

Tier 3 evidence items excluded from all score computations. Stored in DB but not counted.

BR: Badge Assignment

Condition	Badge Sub-type
KLIMA_SCORE ≥ 80	Green
KLIMA_SCORE ≥ 55	Yellow
KLIMA_SCORE ≥ 30	Orange
KLIMA_SCORE < 30	Red
Any pillar null + unrecoverable	Grey GREY_NO_DATA
Questionnaire NO_RESPONSE + votes null	Grey GREY_REFUSED

votes null AND actions < 2 items	Grey GREY_NEW_CANDIDATE
overall_confidence < 0.40	Grey GREY_LOW_CONFIDENCE

Grey sub-type stored in DB. One grey visual on public site. Sub-type text shown in detail view only.

BR: Score Versioning

- Every score record: formula_version, calculated_at, input_snapshot_json, calculated_by
- Re-scoring after any update: new version inserted; previous retained
- All formula constants in scoring_config table — adjustable without code deploy

NON-FUNCTIONAL REQUIREMENTS

NFR: Performance

- Single-candidate recalculation: < 1 second
- All 16 candidates: < 30 seconds bulk recalculation
- Triggered via Supabase DB webhook on pillar score update; idempotent

SC: Implementation

- Supabase Edge Function (TypeScript)
- All constants in scoring_config table: cap values, badge thresholds, confidence floor
- Input validation: reject pillar scores outside 0.0–100.0; log and alert

CAP-08	Manual Evidence & Override System	Phase MVP P0
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GENERAL SECTION

GS: Objective

First-class evidence entry for all pillars — SLOVÁ and SKUTKY. Used when any AI agent is unavailable, fails, or produces output that needs correction. Equal in schema rigour to AI-generated evidence. Not a fallback.

CAP-08 covers all pillars: — SLOVÁ: manually enter program scores or citations when CAP-02 fails; ingest questionnaire responses that arrive outside the submission form; add social media evidence items in Phase 2. — SKUTKY: enter council votes and documented actions when CAP-03 is unavailable (primary path in MVP). The evidence schema is identical regardless of pillar. The scoring engine (CAP-07) routes each evidence item to the correct pillar based on source_type.

GS: MVP Scope

- Single-item evidence entry with full evidence schema including climate_relevance_tier
- Override of AI-generated evidence items
- Physical document exception (scan upload required)

GS: Post-MVP → Phase 2

- Bulk CSV/JSON import

GS: No-URL Exception (tightened)

Permitted ONLY if: (1) physical document scan uploaded to Supabase Storage generating an internal URL, OR (2) second reviewer approves the exception explicitly. A score point with no URL and no uploaded document is unverifiable and legally weak.

USE CASES

UC: Enter Single Evidence Item

36. Reviewer opens candidate; navigates to relevant pillar.
37. Clicks "Pridať dôkaz" (Add evidence).
38. Selects evidence_type from enum — system shows pre-computed point value.
39. Selects climate_relevance_tier: Tier 1 = source document states the environmental connection (paste citation); Tier 2 = environmental connection is real but not stated in source (reviewer must write justification in reviewer_note); Tier 3 = no credible environmental connection (item stored but not scored).
40. Tier 2 prompt shown: "Uvedte dôvod environmentálnej relevantnosti (bude zverejnený ako súčasť citácie)."
41. Tier 3 warning shown: "Táto položka bude vylúčená zo skórovania a vyžaduje schválenie druhého hodnotiteľa."
42. Fills: date, topic, direction (pro/anti/neutral), description, source URL, citation_text ≤280 chars, confidence (1–3 stars).
43. Submits; evidence item created; CAP-07 recalculates; running sub-total updates.

UC: Override AI Evidence

- 44. Reviewer opens AI-generated evidence item.
- 45. Clicks "Override" — original values shown alongside edit fields.
- 46. Enters corrected values + mandatory override_reason.
- 47. Override logged immutably with reviewer_id and original value.
- 48. CAP-07 recalculates using overridden values.

BR: Scoring Rules Enforced in Form

- evidence_type selector maps to predefined point values — reviewer cannot enter points manually
- System computes points from evidence_type + direction — not editable
- For SLOVÁ entries (source_type = program | questionnaire | social_post): point contribution derived from normalized sub-score, not a fixed point value; form shows current sub-score impact
- For SKUTKY entries (source_type = council_vote | resolution | initiative | track_record): point value shown from evidence_type table before submit
- Documented Actions maximum cap enforced at DB constraint level
- Tier field required — form cannot submit without it
- Running pillar sub-total shown in real-time; Tier 3 items shown but excluded from total

Mandatory evidence schema — every score point, every pillar, every source

- source_type: council_vote | resolution | initiative | program | questionnaire | social_post | manual_entry
- url: required; internal storage URL if physical document scanned; N/A only with admin approval + second reviewer
- date_accessed: ISO timestamp
- citation_text: verbatim excerpt ≤280 chars
- confidence: float 0.0–1.0
- climate_relevance_tier: 1 (explicit — source states environmental connection) | 2 (implicit — reviewer documents environmental connection) | 3 (excluded — no credible environmental connection; not scored; requires secondary reviewer to reclassify)
- reviewer_note: required if tier=2 (must document environmental connection), if tier=3, if score adjusted, or if no-URL exception

BR: Additional Data Fields

Field	Description
evidence_items.entered_by	FK to NGO users — always set for manual entries
evidence_items.is_ai_generated	false for all CAP-08 entries
evidence_items.computed_points	Derived from evidence_type + direction; not editable
evidence_items.requires_second_reviewer	true for Tier 3 or no-URL exceptions

NON-FUNCTIONAL REQUIREMENTS

NFR: Usability

- Point value shown before submit — reviewer sees scoring impact

- Running pillar sub-total in real-time
- Tier 3 items visually greyed out in evidence list with "Vylúčené" label

NFR: Audit

- Every entry and override: immutable log with entered_by, timestamp, computed_points, tier
- Manual entries marked "Manuálne" in review dashboard

NFR: Security

- Researcher: can add and submit; cannot approve own entries
- Admin: can approve, override, authorise no-URL exceptions